

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Salinity	WQP 002	1.0	

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1. Title

Standard Operating Procedure for Seawater Salinity Determination

2. Applicability

This SOP applies to the determination of *in situ* seawater salinity at shallow (<30 m) coral reef sites around Puerto Rico.

3. Purpose

- 3.1 Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater salinity profiles at shallow coral reef sites around Puerto Rico.
- 3.2 Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater salinity data following the Quality Assurance Project Plan.

4. Method

- 4.1 Standard Method 2520 B (Baird et al., 2017).
- 4.2 Hydrolab HL4 Multiparameter Sonde is used to measure *in situ* seawater surface, bottom, and depth profiles of salinity.

5. Health and Safety

- 5.1 Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2 When loosening removable parts of a manifold probe, face the probe away from your body and others, as pressure may build up within the instrument resulting in caps, sensors, or other parts that can shoot out and cause injury.

6. General Precautions

- 6.1 Always maintain two points of contact when deploying the HL4 sonde on the boat to avoid losing the instrument overboard.
- 6.2 Ensure the boat engines are idle and the captain approves of the sonde being deployed overboard before deploying the sonde to ensure that the instrument cables do not become entangled with the boat propeller.
- 6.3 During maintenance or replacement of the multiprobe battery, place it horizontally on a planar surface before removing the battery cap to avoid water leaks into the multiprobe battery compartment during maintenance or replacement.

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- 6.4 In case water leaks into the battery compartment, remove the batteries, pour the water out, thoroughly dry the compartment, and install a new battery.
- 6.5 If there are problems with the software, call the manufacturer (800-949-3766) or email natechsupport@otthydromet.com.
- 6.6 HydroLab HL4 Sonde requires that sensors are calibrated in the following order:
 - 6.6.1 Temperature (by manufacturer only) → conductivity → LDO → pH
- 6.7 Do not allow the sensors on the HydroLab HL4 Sonde to dry.
- 6.8 For storage after deployment:
 - 6.8.1 rinse the handheld surveyor, the sonde sensors and the sensor guard with tap water,
 - 6.8.2 rinse the storage cup with tap water,
 - 6.8.3 remove the sensor guard and replace it with the storage cup while it is still wet to allow the sensors to remain humid but **not submerged**.

7. Training

- 7.1 Personnel shall be trained to calibrate and operate the multiparameter HL4 sonde before use.
- 7.2 Personnel shall receive training on how to complete the corresponding forms for this SOP.

8. Required Materials

- 8.1 Hydrolab HL4 Multiparameter Sonde
- 8.2 Hydrolab Handheld Surveyor
- 8.3 Handheld Surveyor USB cable
- 8.4 Mild detergent to clean the sensor
- 8.5 Conductivity standards 35 psu and 72 psu
- 8.6 Warm water to clean the sensor
- 8.7 Soft toothbrush to clean the sensor
- 8.8 Extra “D” battery for replacement
- 8.9 Calibration Form (Appendix I)
- 8.10 Field Form (Appendix F)

9. Procedure

9.1 Calibration

Calibration for the Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) should follow manufacturers instructions (~3 months) to ensure accuracy and control for any potential sensor drift throughout the sampling event.

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9.1.1 Run a calibration check before every sampling event to ensure sensors are working within acceptable ranges.

9.1.2 To run a calibration check:

- 9.1.2.1 Place the HL4 Sonde and handheld Surveyor® near a sink.
- 9.1.2.2 Connect the sonde to the handheld Surveyor® HL and press the **power button** on the handheld.
- 9.1.2.3 Press the **RIGHT arrow** until you read “Sonde_HL4” on the top of the display window.
- 9.1.2.4 Select “*Calibration*” → select “*Conductivity*” → press the **DOWN arrow** and select “*Check Calibration*”.
- 9.1.2.5 Fill the storage cup 1/4 of its capacity with 35 psu calibration standard, screw the storage cup to the sonde, and shake vigorously for at least 6 seconds, empty the storage cup.
- 9.1.2.6 Repeat three times.
- 9.1.2.7 Fill the storage cup 3/4 of its capacity with 35 psu conductivity standard → screw the storage cup to the HL4 sonde → allow for 1 minute for stabilization → write down the calibration standard’s psu in the provided text box → press **OK button** to accept and allow for stabilization.
- 9.1.2.8 Press the **OK button** again on “Keep Calibration Date” to save the calibration check date.

9.2 Calibration

9.2.1 Follow steps 9.1.2.1 - 9.1.2.3 → select “SpCond 2-point Calibration (psu)”

- 9.2.1.1 Fill storage cup 1/4 of its capacity with room temperature distilled water to the storage cup → shake vigorously for 6 seconds → discard water → repeat 3 times total
- 9.2.1.2 Remove the storage cup and completely dry the conductivity sensor by gently blotting it with clean Kimwipes.
 - 9.2.1.2.1. Allow ~ 1 min for stabilization stable at “0.000” press **OK button** → wait for stabilization → press **OK button** to continue.
- 9.2.1.3 Repeat step 9.2.1.1 two additional times but using a calibration standard of 72 psu.
 - 9.2.1.3.1. Before accepting calibration, tilt the HL4 sonde in different directions and gently tap the storage cup with the standard solution to remove any bubbles stuck on the sensors
- 9.2.1.4 Press the OK button on Keep Calibration Check.

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9.2.2 Rinse the storage cup and sensors 3 times using regular tap water → place the tap water-humid storage cup on the sonde until it is ready to be deployed.

9.3 Prepare the sonde for deployment.

9.3.1 Connect the sonde to the handheld Surveyor® HL using the deployment cable and press the **power button** on the handheld.

9.3.2 After checking calibration for each sensor, attach the tap water-humid storage cup to the sonde until arrival to the sampling site.

9.3.3 Once at the sampling site, replace the storage cup with the sensor guard for deployment.

9.4 Sample Collection

9.4.1 While the HL4 sonde is connected to the handheld and deployment cable, press the **power button** → **RIGHT arrow** until you read “Sonde_HL4”.

9.4.1.1 Select “Overview” and make sure all sensors have started properly (there should be a check mark to the right of each parameter (e.g. Conductivity ✓).

9.4.1.1.1. Once checked, press the **return button** ↵.

9.4.2 Without placing the HL4 sonde into the water, and while on the “Sonde_HL4” menu, select “Real-time” → press **OK button** to start data save.

9.4.2.1 Press the **DOWN arrow button** once and then the **UP arrow button** once to select “New Log” → press **OK button**

9.4.2.1.1. Replicate casts should be saved as “SiteCode_R2” instead of the standard log name. **Otherwise, leave the default log name.**

9.4.2.2 Select “Location” and select the corresponding “Site Code” → press **OK button**.

9.4.2.3 Press the **RIGHT arrow button** to save the settings → press **OK button** to begin data recording (you should see the number of “Measurements” on the lower left corner of the display window steadily increasing).

9.4.3 Gently lower the HL4 Sonde ~0.5 m into the water → using Sampling Log (Appendix F) record the desired parameters and their respective depths, **leaving surface temperature for last** to allow for stabilization of the sensor.

9.4.4 Lower the sonde at a slow and steady rate until it gets ~ 0.5m from the bottom with caution not to touch the bottom and stir sediment up.

9.4.5 Record the bottom salinity (psu) and corresponding depth (m) reading on the Sampling Log (Appendix F).

9.4.6 Pull the sonde back up into the boat.

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9.4.7 Press **OK button** to stop recording the cast → press **return button** ↵ → turn off the handheld surveyor.

9.4.8 If other sites are to be sampled, place the Hydrolab sonde on a bucket that holds clean tap water until arrival at the next site.

9.4.8.1 If no other sites are to be sampled, follow **step 6.8 for storage after deployment**.

9.5 Transfer data from handheld Surveyor to PC

9.5.1 Connect the surveyor to the PC using the manufacturer's USB cable and press the **power button** on the handheld.

9.5.2 Open the *Hydrolab Operating Software* → select the handheld surveyor from the available devices list → click on “Overview” on the navigation bar.

9.5.2.1 On the lower-right of the display, select the log files you wish to transfer and click “Transfer...”. → select save folder → click “Save” (these will be in Hydrolab File Format).

9.5.2.2 Open the Hydrolab File Format saved files → click “Export...” → click **OK** → rename the file as: *SiteCode_Date of collection_default log name* (e.g., BOYA20_12MAY23_Log_12345_6789) → select format type (e.g., CSV, XLSX) → click “Save”.

10. Documentation

10.1 Field data will be recorded in Field Form (Appendix F).

10.2 Calibration data will be recorded in Calibration Form (Appendix I).

10.3 Field and Calibration forms shall be kept in the designated file cabinet at the CCRI office.

10.4 Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.

10.5 All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.

10.6 All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

11.1 After each sampling event, rinse the handheld surveyor, the sensors, the storage cup, and the sensor guard thoroughly with clean warm fresh water.

11.2 After each sampling event, inspect sensors carefully for physical damage.

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- 11.3 In case of unwanted material deposits over the sensors, gently clean sensors with a soft brush and mild detergent. Do not use dishwashing liquid or detergents with skin moisturizer as these can create films over the probes.
- 11.4 For long-term storage (>24hr), rinse the storage cup and sensors with tap water and screw the storage cup on the sensors while it is still humid. **No sensor should be permanently touching or be submerged in water.**
 - 11.4.1 Do not fill the storage cup with distilled water as this may damage the probes.
- 11.5 Replace sonde “D” battery as necessary.
 - 11.5.1 The handheld surveyor “Overview” shows when the battery charge is low.
- 11.6 Perform a monthly preventive maintenance.
 - 11.6.1 Carefully inspect sensors, O-rings, connective pins, and physical integrity of the sonde in general.
 - 11.6.1.1 Refer to HL Series Sensors DOC026.97.80349 document for instructions.
 - 11.6.2 Lubricate or replace O-rings, if necessary.
 - 11.6.3 Follow manufacturer indications for sensor maintenance.
 - 11.6.3.1 Refer to “HL Series Sonde DOC026.97.80348” document for instructions.

12. Troubleshooting

- 12.1 In case of reading errors, check the sensor path for bubbles, grease, or other contamination.
- 12.2 In case of unstable readings, contact the manufacturer to replace the sensor.

13. QA/QC

- 13.1 As per QAPP Table 4, seawater salinity should be measured in practical salinity units (psu) with an accuracy of ± 0.50 psu ($\pm 20\%$) and a precision of $< 20\%$.
- 13.2 In accordance with the QAPP, 10% of all Hydrolab casts will be repeated to ensure precision of replicate surface and bottom measurements is $< 20\%$.
- 13.3 All field samples should adhere to typical seasonal salinity values observed in coastal waters of Puerto Rico.
- 13.4 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

14. References

- 14.1 Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.

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- 14.2 Hydrolab (2014). Hydrolab Operating Software. DOC026.53.80346I
- 14.3 Hydrolab (2017). HL Series Sonde User Manual. DOC026.97.80348
- 14.4 Hydrolab (2019). HL Series Sensors User Manual DOC026.97.80349